



It is designed to provide security analysts with real-time situational awareness by integrating telemetry from Wazuh, Sysmon, and the custom ML engine through Kibana and Vega.

The dashboard is structured into several functional modules:

1. Real-Time Threat Metrics

- **Threat Overview:** Tracks the total volume of ingested alerts and filters them into high-risk events for immediate analyst review.
- **Severity Distribution:** Provides a categorical breakdown of alerts ranging from Low to Critical to assist in incident prioritisation.
- **Live Agent Status:** Monitors the activity and connection health of various endpoints (e.g., Windows 10 targets) within the environment.

2. ML Risk & Attack Progression

- **Dynamic Risk Gauges:** Visualises the live Current Threat Score, derived from the combination of ML base probability and contextual behavioral analysis.

- **Attack Progression Timeline:** A custom Vega-based visualization that tracks risk fluctuations over time. It includes a defined 90% response threshold line to show when an attack chain is nearing automated containment.
- **ML Scoring Breakdown:** Displays the ML BASE, FINAL RISK, and SESSION RISK metrics to show how the system correlates isolated events into a cumulative threat score.

3. MITRE ATT&CK & Technique Correlation

- **Technique Correlation Matrix:** Maps detected security events to specific MITRE ATT&CK techniques (e.g., T1110 for Brute Force, T1550.002 for Lateral Movement), providing a standardised view of attacker tactics.
- **Attack Stage Classification:** Groups alerts into logical phases such as Reconnaissance, Persistence, and Privilege Escalation.

4. Automated Response & Forensics

- **Active Response Triggering:** Captures the exact moment the automated containment is activated, detailing the Attacker IP, Target Host, and the specific attack stage that triggered the block.
- **Blocked IP Activity:** Maintains a live table of IPs successfully contained via pfSense integration, including timestamps of the enforcement actions.
- **Forensic Report Integration:** Features a dedicated section for Incident Reporting, providing direct browser-based access to PDF reports generated by ReportLab.

5. Infrastructure & System Health

- **Security Stack Status:** Monitors the operational status of the underlying security components, including the Wazuh Manager, Elastic Stack, and ML Correlation Engine.
- **Response Linkage:** Confirms secure SSH connectivity to the pfSense firewall, ensuring the automated response pipeline is active.

This dashboard serves as the final integration point for the Hybrid Detection and Response System, transforming complex JSON logs into actionable security intelligence.